Question 4

(a)

There is an increase in separation hence an increase in the molecular potential energy. **B1**

Temperature remains constant so the microscopic kinetic energy remains constant. **B1**

(b)(i)

$n = pV / RT$

$= (1.2 \times 10^5 \times 0.24) / [8.31 \times (273.15 + 17)]$

$= 12$ mol **A1**

Marker's comments

(b)(ii)

change	work done on gas/kJ	thermal energy supplied to gas /kJ	increase in internal energy of gas/ kJ
AB	19.2	–48.0	–28.8
BC	0	28.8	28.8
CA	–31.6	31.6	0

Work done AB = $1.2 \times 10^5 \times (0.24 - 0.08) = 19200$ J = 19.2 kJ

Work done BC = 0 (since no change in volume)

Work done CA= –31.6 (value given in question, students to take note that the work done is negative since CA is an expansion process)

[1 mark for work done]

CA: increase in internal energy = 0 (since no change in temperature) and thermal energy supplied = 31.6 (by first law of thermodynamics)

[1 mark for CA]

Increase in internal energy of AB = -28.8, increase in internal energy of BC = 28.8

Change in internal energy (AB + BC + CA) = 0

[1 mark for increase internal energy and thermal energy of BC]